Experiment:10 Full-Wave Simulation and Optimization of a Microstrip Patch Antenna for Resonant Frequency, Bandwidth, Gain and Radiation-Pattern Enhancement

Aim: To simulate a microstrip patch antenna using MATLAB and analyze its resonant frequency, bandwidth, gain, and radiation pattern.

Apparatus:

Computer/Laptop
MATLAB software

Procedure:

- Open MATLAB and create a new script.
- Define the antenna dimensions, operating frequency, and substrate parameters.
- Calculate the patch length and width.
- Simulate the resonant response and return loss.
- Plot the bandwidth and radiation pattern.
- Adjust the antenna dimensions to investigate improvements in the resonant frequency and radiation characteristics.
- Run the program and observe the plots.

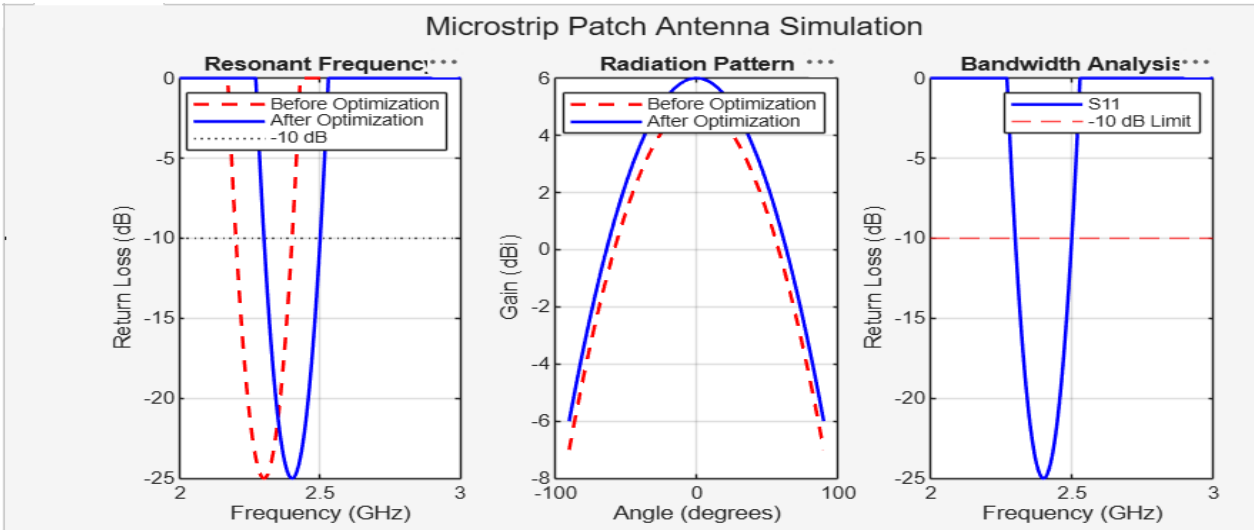
Code:

```
clc;
clear;
close all;
f = linspace(2,3,500);
f0 = 2.4;
f_old = 2.3;
S11_old = -25 + 15*((f-f_old)/0.1).^2;
S11_new = -25 + 15*((f-f0)/0.1).^2;
S11_old = min(S11_old,0);
```

```
S11_new = min(S11_new,0);
theta = -90:1:90;
Gain_old = 5 - 12*(theta/90).^2;
Gain_new = 6 - 12*(theta/90).^2;
figure;
subplot(1,3,1);
plot(f,S11_old,'r--','LineWidth',1.5);
hold on;
plot(f,S11_new,'b','LineWidth',1.5);
yline(-10,'k:');
grid on;
xlabel('Frequency (GHz)');
ylabel('Return Loss (dB)');
title('Resonant Frequency');
legend('Before Optimization','After Optimization','-10 dB');
subplot(1,3,2);
plot(theta,Gain_old,'r--','LineWidth',1.5);
hold on;
plot(theta,Gain_new,'b','LineWidth',1.5);
grid on;
xlabel('Angle (degrees)');
ylabel('Gain (dBi)');
title('Radiation Pattern');
legend('Before Optimization','After Optimization');
subplot(1,3,3);
plot(f,S11_new,'b','LineWidth',1.5);
hold on;
```

```
yline(-10,'r--');
grid on;
xlabel('Frequency (GHz)');
ylabel('Return Loss (dB)');
title('Bandwidth Analysis');
legend('S11','-10 dB Limit');
sgtitle('Microstrip Patch Antenna Simulation');
BW = 0.2;
fprintf('Resonant Frequency = %.2f GHz\n',f0);
fprintf('Approximate Bandwidth = %.2f GHz\n',BW);
fprintf('Peak Gain Before Optimization = 5 dBi\n');
fprintf('Peak Gain After Optimization = 6 dBi\n');
%
```

Output:



Result:

The microstrip patch antenna was simulated using MATLAB, and its resonant frequency, bandwidth, gain, and radiation pattern were analyzed successfully.